

Conditional Statements + Loops

Concept	Purpose	Example
if	Decision making	if age > 18
elif	Checks multiple conditions	elif marks >= 70
else	Executes when no condition matches	else:
for loop	Iterates over sequence	for i in range(5)
while loop	Runs while condition is true	while x < 10
break	Stops loop immediately	break
continue	Skips current iteration	continue
pass	Placeholder statement	pass

Conditional Statements

```
In [2]: x = 8
r = x%2
if r == 0:
    print('Even Number')

print('Odd Number')
```

Even Number
Odd Number

```
In [3]: x = 8
r = x%2
if r == 0:
    print('Even Number')
else:
    print('Odd Number')
```

Even Number

```
In [4]: x = 9
r = x%2
if r == 0:
    print('Even Number')
else:
    print('Odd Number')
```

Odd Number

```
In [6]: x = 9
r = x%2

if r == 0:
    print('Even Number')
elif r == 1:
    print('Odd Number')
else:
    print('Not a number')
```

Odd Number

```
In [8]: x = 9
r = x%2

if r == 0:
    print('Even Number')
    if x > 5:
        print("Greater than 5")
elif r == 1:
    print('Odd Number')
    if x > 5:
        print("Greater than 5")
else:
    print('Not a number')
```

Odd Number
Greater than 5

Loops

```
In [9]: for i in range(10):
        print(i)
```

0
1
2
3
4
5
6
7
8
9

```
In [10]: fruits = ["Apple", "Banana", "Mango"]

for val in fruits:
    print(val)
```

Apple
Banana
Mango

```
In [12]: count = 1
while count < 5 :
```

```
print(count)
count+= 1
```

1
2
3
4

```
In [13]: for i in range(20):
          if i == 5:
              break;
          print(i)
```

0
1
2
3
4

```
In [14]: for i in range(20):
          if i == 5:
              continue;
          print(i)
```

0
1
2
3
4
6
7
8
9
10
11
12
13
14
15
16
17
18
19

```
In [15]: for i in range(20):
          pass
```

Print Even Numbers till 50

```
In [16]: for i in range(50):
          if i%2 == 0:
              print(f"{i} is Even Number")
```

0 is Even Number
2 is Even Number
4 is Even Number
6 is Even Number
8 is Even Number
10 is Even Number
12 is Even Number
14 is Even Number
16 is Even Number
18 is Even Number
20 is Even Number
22 is Even Number
24 is Even Number
26 is Even Number
28 is Even Number
30 is Even Number
32 is Even Number
34 is Even Number
36 is Even Number
38 is Even Number
40 is Even Number
42 is Even Number
44 is Even Number
46 is Even Number
48 is Even Number

Sum Positive Numbers

```
In [17]: numbers = [5, -2, 10, -1, 7]
sum_pos = 0
for n in numbers:
    if n > 0:
        sum_pos += n

print(f"Sum of positive numbers {sum_pos}")
```

Sum of positive numbers 22

Loop with else

The else block executes if loop ends naturally.

```
In [18]: for i in range(5):
print(i)
else:
    print("Loop Ends")
```

0
1
2
3
4
Loop Ends

else Does NOT Execute After break

```
In [23]: for i in range(5):
          print(i)
          if i==3:
              break
      else:
          print("Loop Ends")
```

```
0
1
2
3
```

Print numbers divisible by 3 from 1–30

```
In [28]: for i in range(1,31):
          if i%3 == 0:
              print(i)
```

```
3
6
9
12
15
18
21
24
27
30
```

Find largest number in a list

```
In [31]: l = [2,3,5,819,3,4]
          largest = l[0]

          for i in l:
              if largest < i:
                  largest = i
          print(f"Largest number {largest}")
```

Largest number 819

Print star pattern

```
In [40]: for i in range(1,6):
          print('*'*i)
```

```
*
**
***
****
*****
```

Check palindrome number

```
In [48]: word = input('Enter a string to check palindrome')
palindrome=''
length = len(word)-1

for i in range(len(word)):
    palindrome+= word[length-i]
if palindrome == word:
    print("Given String is a palindrome")
else:
    print("Given String is not a palindrome")
```

Given String is a palindrome

Count vowels in a string

```
In [56]: string = input('Enter a string to count of vowels').lower()
vowels=['a','e','i','o','u']
count = 0

for i in string:
    if i in vowels:
        count+=1

print(f"Count of vowels in {string} is {count}")
```

Count of vowels in avinash is 3

In []: